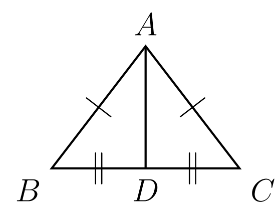
Completing Two-Column Triangle-Congruence Proofs

High School Geometry

Focus: SSS and SAS proofs, ASA/AAS/HL proofs, and CPCTC equations.

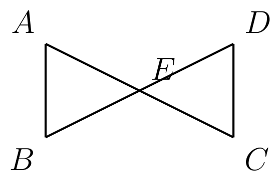
Directions: Fill in every empty cell. A statement must be a congruence or a fact about the figure; a reason must be a given, a definition, a theorem, or a postulate — never "it looks like it". The last line of a proof always names the test used: SSS, SAS, ASA, AAS or HL. Marks on a figure repeat the givens; they are not extra information.

1. **Given:** $\overline{AB} \cong \overline{AC}$; D is the midpoint of $\overline{BC}$. **Prove:** $\triangle ABD \cong \triangle ACD$.



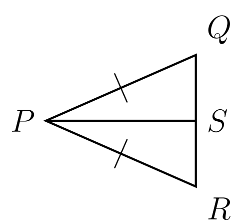
Statements	Reasons
1. $\overline{AB} \cong \overline{AC}$	1. Given
2. D is the midpoint of $\overline{BC}$	2. Given
3. $\overline{BD} \cong \overline{DC}$	3. Definition of midpoint
4. $\overline{AD} \cong \overline{AD}$	4.
5. $\triangle ABD \cong \triangle ACD$	5.

2. Given: E is the midpoint of $\overline{AC}$ and of $\overline{BD}$. **Prove:** $\triangle AEB \cong \triangle CED$. (Write every reason.)



Statements	**Reasons**
1. $\overline{AE} \cong \overline{EC}$	1.
2. $\angle AEB \cong \angle CED$	2.
3. $\overline{BE} \cong \overline{ED}$	3.
4. $\triangle AEB \cong \triangle CED$	4.

3. Given: $\overline{PQ} \cong \overline{PR}$; $\overrightarrow{PS}$ bisects $\angle QPR$. **Prove:** $\triangle PQS \cong \triangle PRS$. (Fill in the missing statements *and* reasons.)

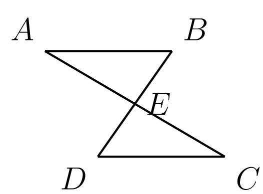


Statements	**Reasons**
1. $\overline{PQ} \cong \overline{PR}$	1. Given
2. $\overrightarrow{PS}$ bisects $\angle QPR$	2. Given
3.	3. Definition of angle bisector
4. $\overline{PS} \cong \overline{PS}$	4.
5.	5.

4. $\triangle ABC \cong \triangle DEF$. On the corresponding sides, $AB = 3x - 4$ and $DE = 2x + 5$. Corresponding parts of congruent triangles are congruent, so these two lengths are equal. Write the equation this gives and solve it for x.

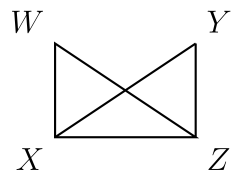
Answer: ____________________

5. Given: $\overline{AB} \parallel \overline{DC}$; E is the midpoint of $\overline{AC}$; B, E, D are collinear. **Prove:** $\triangle ABE \cong \triangle CDE$.



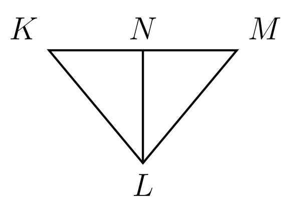
Statements	**Reasons**
1. $\overline{AB} \parallel \overline{DC}$	1. Given
2. $\angle BAE \cong \angle DCE$	2.
3.	3. Definition of midpoint
4. $\angle AEB \cong \angle CED$	4.
5.	5.

6. Given: $\angle W \cong \angle Y$; $\angle WXZ \cong \angle YZX$. **Prove:** $\triangle WXZ \cong \triangle YZX$. (The reasons are printed — write the statements.)



Statements	Reasons
1.	1. Given
2.	2. Given
3.	3. Reflexive Property of Congruence
4.	4. AAS

7. Given: $\overrightarrow{LN}$ bisects $\angle KLM$; $\angle K \cong \angle M$. **Prove:** $\triangle KLN \cong \triangle MLN$. (Write the whole proof.)



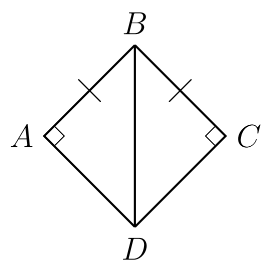
Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.

8. $\triangle ABC \cong \triangle DEF$, and the corresponding angles satisfy $m\angle A = 5x + 12$ and $m\angle D = 7x - 8$.

Use CPCTC to write an equation in x and solve it. (Then check: what is $m\angle A$?)

Answer: ____________________

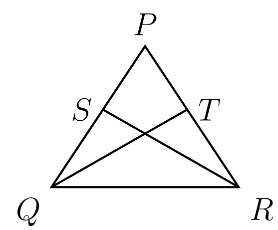
9. Given: $\angle A$ and $\angle C$ are right angles; $\overline{AB} \cong \overline{CB}$. **Prove:** $\triangle ABD \cong \triangle CBD$. Then, underneath, explain in one or two sentences why SSA is not a valid test in general but HL is allowed here.



Statements	**Reasons**
1.	1.
2.	2.
3.	3.
4.	4.

10. Challenge (overlapping triangles). Given: S lies on $\overline{PQ}$, T lies on $\overline{PR}$, $\overline{PS} \cong \overline{PT}$, and $\overline{PQ} \cong \overline{PR}$. **Prove:** $\triangle PQT \cong \triangle PRS$, and then state what CPCTC lets you conclude about $\overline{QT}$ and $\overline{RS}$.

Hint: redraw the two triangles separately first — they share $\angle P$.



Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.